Grass Orientation from Drone Images for Wind Patterns

Francesca Mnenula

Dr. Ivan Ramler, Advisor

Data Science & Economics Majors

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Overview

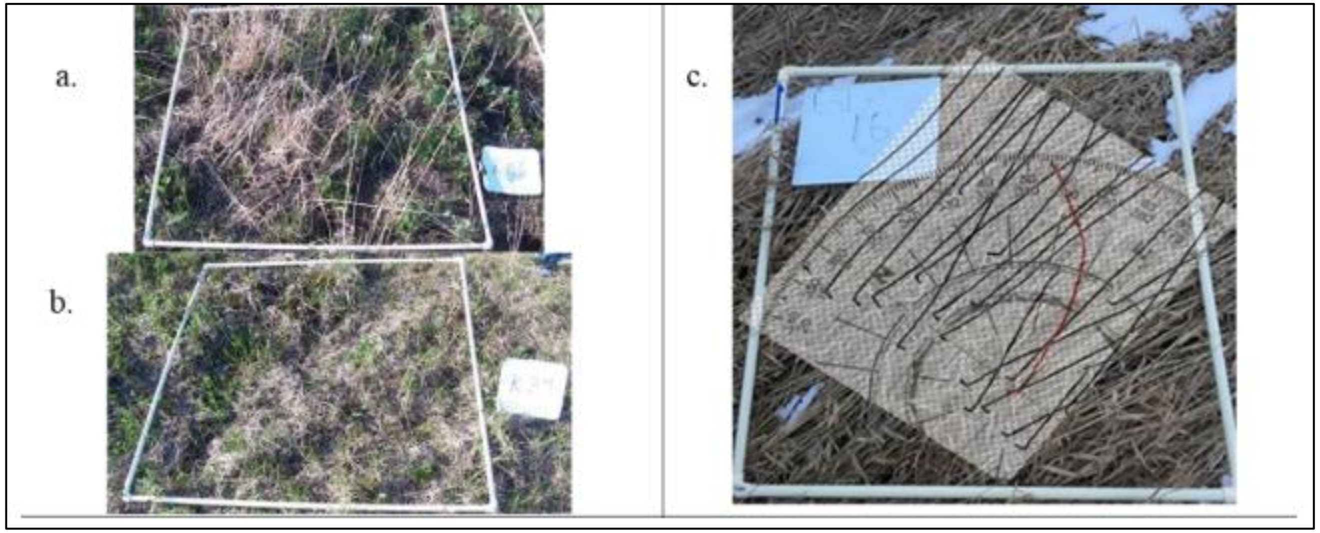
- Find the connection between grass orientation and wind direction
- Use Living Lab tundra grass drone images and ML techniques to retrieve dominant grass orientation
- The technique could be extrapolated to tundra grass images in Alaska

Motivation

- Dr. Rosales' larger study of recalibrating TEK in Arctic regions
- Harsh Arctic climate and wildlife make it difficult to keep weather equipment
- Culturally, residents traditionally use grass orientation to interpret wind direction
- Understanding changes in wind direction is essential for hunting practices

Goal

- Automate a manual process that measured individual blades of grass which may have been time consuming



Questions

- Can we use data science techniques to automate this process as well as the human manual process?
- How can we use grass images taken by drones to recalibrate evolving wind direction data?

Stages of Research

1st Step

Can we measure grass orientation from images? (Sunshine, 2024)

2nd Step

Gradient analysis to recover dominant grass orientations from SLU Living Lab images

3rd Step

Alignment between gradient analysis and human-assessment of SLU Living Lab images

4th Step

Alignment between grass orientation and wind direction data

How does the HOG Algorithm work?

```
$ :'data.frame':        60000 obs. of  3 variables:
 ..$ mag   : num [1:60000] 0.519 0.342 0.356 0.408 0....
 ..$ theta : num [1:60000] 130.887 0.991 1.595 9.017 ...
 ..$ radian: num [1:60000] 2.2844 0.0173 0.0278 0.157...
$ :'data.frame':        30000 obs. of  3 variables:
 ..$ mag   : num [1:30000] 0.972 0.613 0.626 0.581 0....
 ..$ theta : num [1:30000] 135 168 176 168 149 ...
 ..$ radian: num [1:30000] 2.35 2.94 3.07 2.93 2.6 ...
```

Why do gradients work for grass images?

- Gradient extraction techniques identify dominant directions and magnitudes, and grass images have *defined, fine, and linear textures*
- Because we are interested in capturing that general trend or flow in the grass, gradient analysis is useful for this project

Determining dominant grass direction

Using gradient analysis techniques:

A raw grass image

theta	mag	radian
123	0.53	2.14
89	0.18	1.55
74	0.11	1.29
90	0.51	1.57
127	0.43	2.21
103	0.44	1.80

Angles and magnitudes detected from this raw grass image

This arrow indicates the circular mean of the angle (theta) values in the gradients data

Determining dominant grass direction

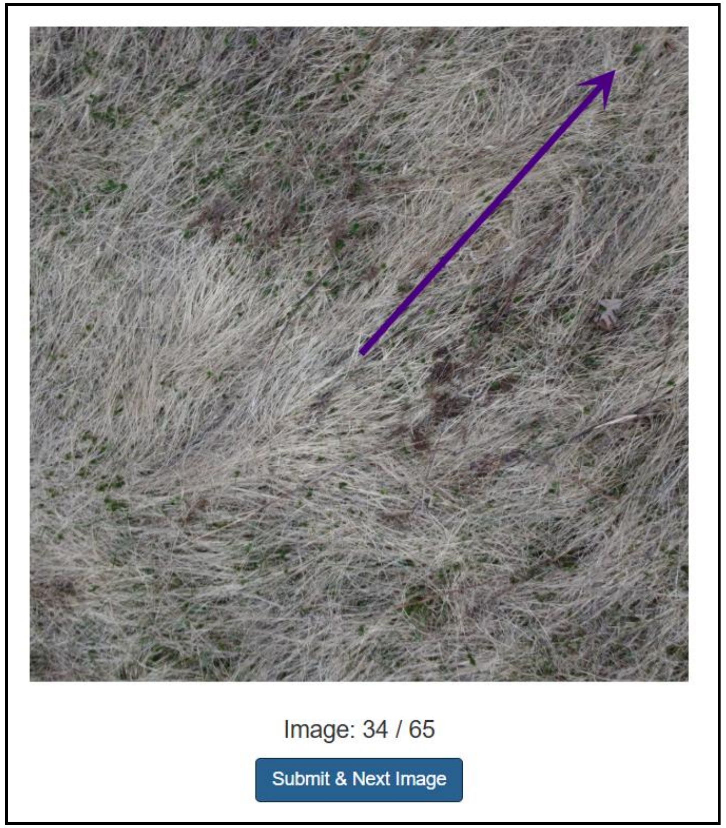
Creating a R Shiny Web App to collect human-assessment data:

- R Shiny is a tool used to build interactive web applications and dashboards
- Displayed the 65 grass images in random order to volunteers and asked them to place an arrow in the direction of the dominant grass direction
- Randomly oriented the images to remove bias of northeast winds
- Collected the arrow’s angular direction and arrow length in a google sheet

Determining dominant grass direction

Using human-assessment of images:

- 45 volunteers placed the arrow in the direction of the predominant grass leaning
- The circular mean of all direction angles submitted by volunteers was calculated



The interface of the R Shiny Web App

Images Taken From SLU Living Lab

Image 1

Image 2

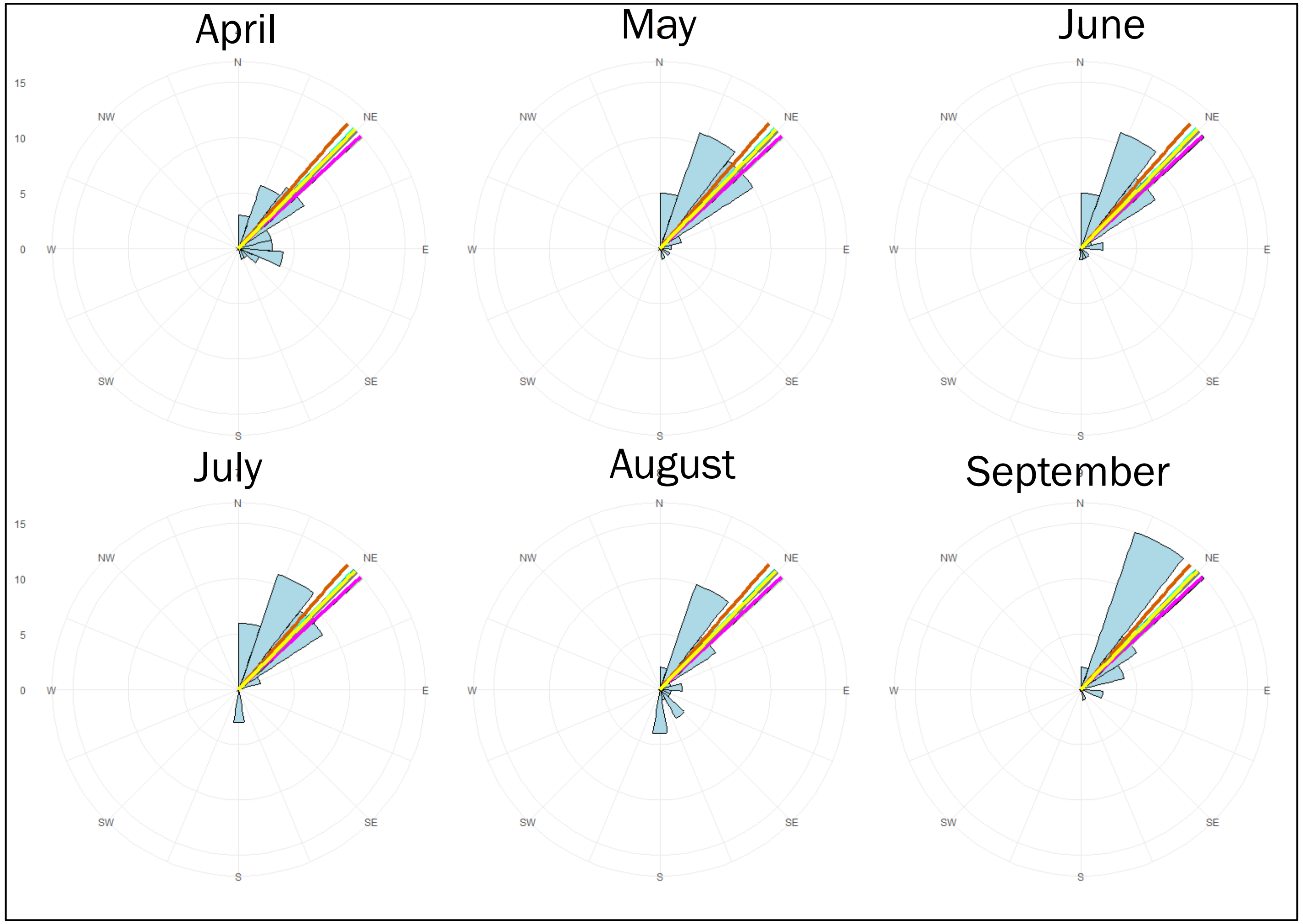
Image 3

Wind Data Collection

- At the SLU Living Lab, Dr. Rosales installed a wind station next to the tundra plants
- Various wind data such as speed and direction is collected every 15 minutes
- Condensed the data to daily averages
- Allows us to compare wind data to grass directions

Wind Direction and Gradient Analysis Grass Direction

Lines of dominant grass directions from 8 different images taken in April 2023



Summary

- Based on several images within our dataset, crowdsourced human-assessment and gradient analysis of images closely align
- The wind direction and grass orientation align best in the Spring, the growing season for the tundra grass
- However, much work remains to be done

Next Steps

- Evaluate what is going on when human-assessment and gradient analysis do not agree
- Guidelines for capturing drone images
- Observe consistency of pattern with 2024 grass images and 2023 wind data




Extensions of Research

A web app where small-holder farmers can upload images to estimate localized wind direction to make decisions (e.g. placing wind blocks)

Use the R Shiny App to crowdsource data like we did for human-assessment of the grass direction

Thank you

Additional Credits to:

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Francesca Mnenula

fmnen22@stlawu.edu

Any questions?